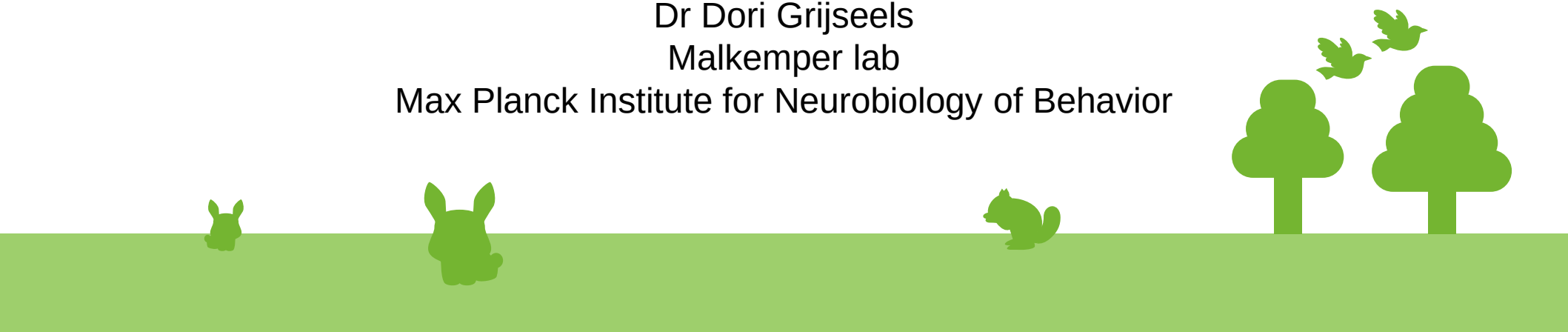
QR Code

Computational tools for bioacoustic research

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Goal of the workshop

- Understand different steps of bioacoustics analysis pipeline
- Have an overview of available tools and resources for further reading
- Able to run a basic pipeline on bioacoustics data using Python



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What it is *not*

- Coding or Python lesson
- Tutorial on one particular tool
- Definitive or only way to do bioacoustics analysis



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Accompanying wiki and code available here:

https://github.com/DoriMG/AnimalBehaviourLiveWorkshop



Schedule for workshop

- Theoretical introduction of topics (10 mins)
- Follow-along practical guide (30-35 mins)
- Questions (10-15 mins)



Resources for the workshop



This presentation



GitHub wiki



3 Jupyter notebooks

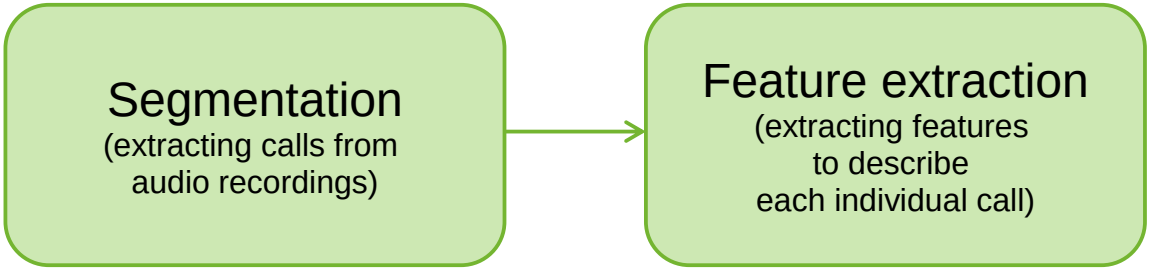


Analysis pipeline

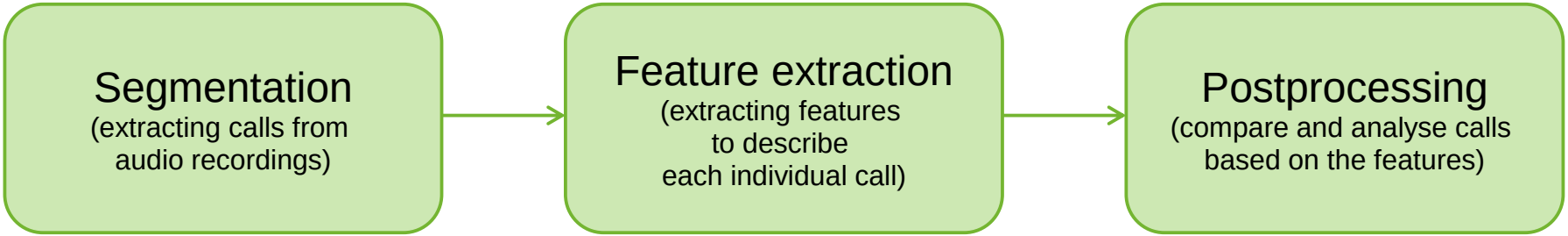
Segmentation
(extracting calls from audio recordings)



Analysis pipeline



Analysis pipeline



Segmentation

- Detect the relevant signals in longer recordings
- Can be done by hand or automated (usually with training on hand annotated data)



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- Avisoft SASLab Pro (paid, free lite version)
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Model architectures (Stowell, 2022):

- ResNet
- VGG
- Custom CNN



Segmentation tools

Package	Language	Pros	Cons
DeepSqueak (Coffey, 2019)	MATLAB	Intuitive GUI	MATLAB-based (paid) Specialized in USVs
ANIMAL-SPOT (Bergler, 2022)	Python	Tested on several species	Setup can be tricky
BirdNET (Kahl, 2021)	Python	Actively maintained and updated Large number of bird species	Specialized for bird classification, potentially less useful for other species



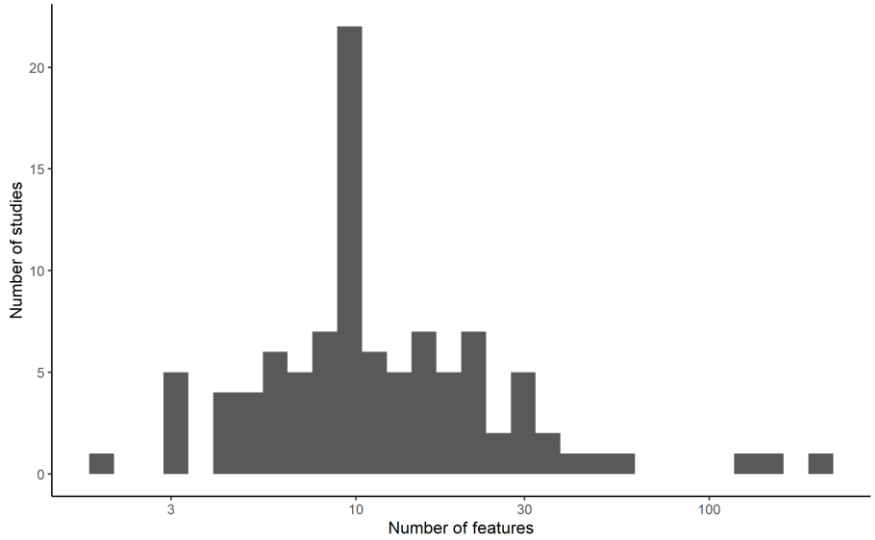
Feature extraction

- Describe sound in a small set of numbers
- Manual extraction vs latent extraction



Feature extraction

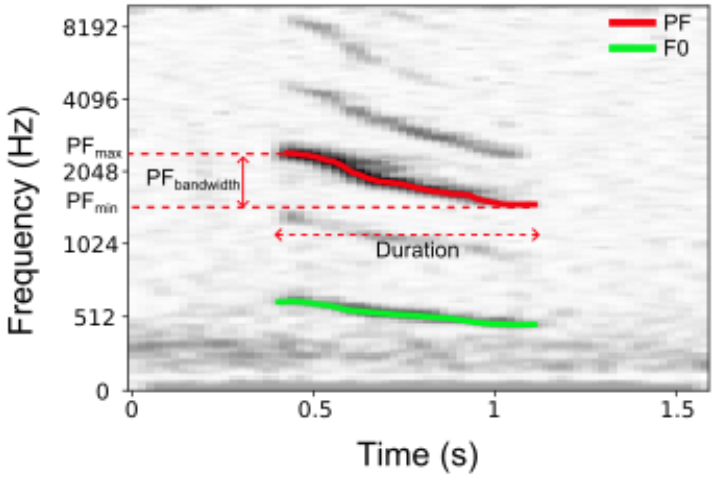
- Describe sound in a small set of numbers
- Manual extraction vs latent extraction



Grijseels et al (2026), Current state of bioacoustics research of vocal repertoires, bioRxiv

Non-exhaustive list of manual features

- Temporal (e.g. duration, time to maximum freq)
- Contour measures (F0, peak frequency)
- Spectrographic (e.g. minimum, maximum, center or bandwidth of frequency)
- Multi-unit measures (e.g. unit duration, number of units, inter unit interval)
- Other (e.g. entropy, amplitude, harmonics-to-noise ratio)



Grijseels et al (2026), Current state of bioacoustics research of vocal repertoires, bioRxiv

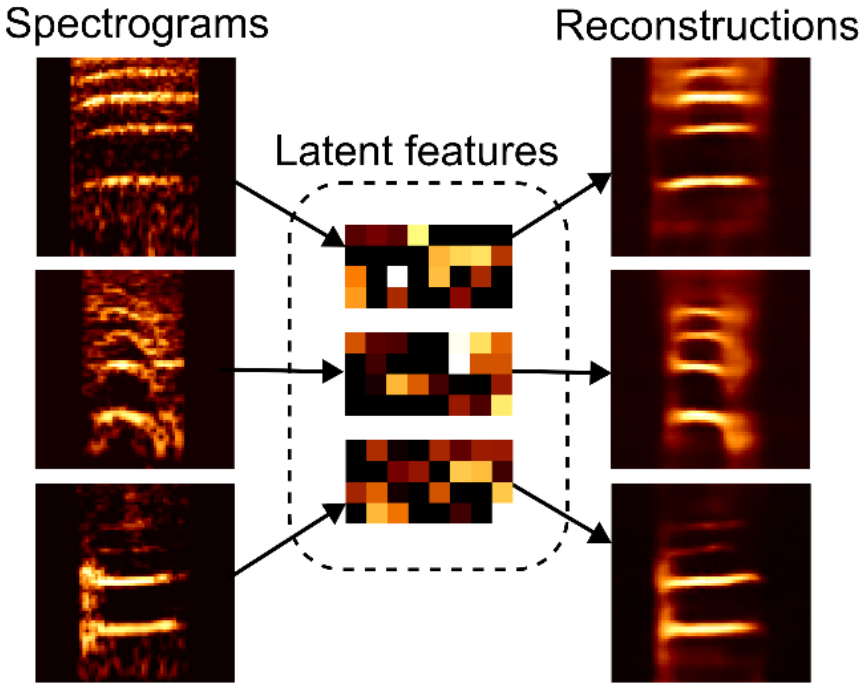
Latent features

- Infer features from raw data using dimensionality reduction methods
- Features do not clearly represent a single measurable feature (e.g. frequency)
- Several ways to do this, including autoencoder and DNN
- Resources:
  - Sainburg et al. (2020) – great overview paper
  - Thomas et al. (2022) – practical guide based on Sainburg paper





Latent features: autoencoder



- Input spectrogram as images of set size
- Pass through convolutional layers to downsample to the bottleneck
- Pass through reverse layers to upsample to original size
- Compare original to reconstruction and update weights
- After training, the bottleneck values are the latent features

Post-processing

Clustering

- Goal: find what calls are most similar to each other
- Many different methods available (see Wadewitz (2015) for discussion)

Classification

- Determine whether vocalizations hold information about certain feature
- Predicted feature flexible (e.g. call type, caller ID, etc)
- Again, many different methods exist



Practical workshop

